

BIOLOGY: MOST REPEATED QUESTIONS

Analysis of Board Paper Sets 57/1/1, 57/5/1, 57/7/1

By Pragya Thunderstudy

1. Sewage Treatment Plant (STP) & BOD

High-Frequency Diagram

Question Type: Diagram/Flow Chart Analysis

- Study the flow chart of the STP and identify 'Primary Treatment' and 'Secondary/Biological Treatment'.
- Why is the primary effluent passed into large aeration tanks?
- Define "**Flocs**" and "**Activated Sludge**".
- Explain how **BOD** is reduced during the process and its significance.

2. Restriction Enzymes (Naming & Action)

Biotech Core

Question Type: Reasoning/Application

- Explain the convention for naming the restriction enzyme **EcoRI**.
- Draw an illustration showing the action of *EcoRI* on DNA (Palindromic sequence & sticky ends).
- How do restriction enzymes facilitate recombinant DNA formation?

3. Population Interactions (Mutualism & Co-evolution)

Ecology Core

Question Type: Reasoning/Case-based

- Explain the **Fig tree and Wasp** interaction as an example of Mutualism.
- Explain "**Sexual Deceit**" in the orchid *Ophrys*.
- Explain how these relationships demonstrate **Co-evolution**.

4. Contraception (Pills & IUDs)

3-Mark Standard

Question Type: Theoretical

- Explain the mode of action of **Oral Contraceptive Pills**.
- Function of **IUDs** like Copper-T or LNG-20 (Phagocytosis/motility suppression).

5. Genetic Disorders (Chromosomal & Mendelian)

Karyotype/Pedigree

Question Type: Diagram Analysis/Case Study

- **Down's Syndrome:** Identify from karyotype (Trisomy 21), cause, and symptoms.
- Differentiate **Turner's** and **Klinefelter's** (makeup and symptoms).
- **Pedigree Analysis:** Determine genotypes for Sex-linked recessive traits.

6. Transgenic Animals & Bio-Ethics

Direct Theory

Question Type: Short Answer

- Define Transgenic Animals.
- Benefits: **Vaccine Safety** (mice), **Biological Products** (Rosie), **Chemical Safety**.

7. Human Reproduction (Diagrams & Cycles)

High-Weightage Diagram

Question Type: Diagram/Graph Analysis

- **Menstrual Cycle:** Study the hormonal levels graph and identify phases.
- **Sperm:** Labelled diagram; explain Acrosome and Mitochondria functions.
- **Seminiferous Tubule:** Identify Spermatogonia, Sertoli cells, and Leydig cells.

8. Biodiversity Conservation

Short Answer favourite

Question Type: Short Answer

- Explain "**The Evil Quartet**" (Habitat loss & Alien Species invasion examples).
- Define **Biodiversity Hotspots** and name India's hotspots.

9. Assisted Reproductive Technologies (ART)

Acronym/Reasoning

Question Type: Acronym Expansion/Reasoning

- Explain procedures: **IVF**, **ZIFT**, **GIFT**, **ICSI**.
- Conditions for recommending **GIFT vs ICSI**.
- Definitions and misuse of **Amniocentesis**.

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Source Analysis complete